

EXAM 7 – SPRING 2011

21. (4 points)

Given the following information for an insurance company:

- Current firm value is \$950 million.
- Underwriting income for the year ended December 31, 2010 was \$40 million.
- Invested assets of \$200 million yielded a total return of 5% in 2010.
- Surplus at December 31, 2010 was \$130 million.
- Risk free interest rate is 3% per annum.

a. (1.5 points)

Calculate the probability of the firm entering a distressed state.

b. (1.5 points)

Discuss three issues that motivate insurers to adopt risk management to avoid financial distress.

c. (1 point)

Discuss two reasons it is difficult for distressed firms to raise bonds.

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Sample Solutions
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Solution 1

a)

E = expected annual earnings

M = multiplier

E=UW Earnings + Investment Earnings

$$=40 + .05 \times 200=50$$

V = expected annual earnings $\times$ multiplier = 950

$$= E \times M = 50 \times M$$

$$V=950 \quad 950=50 \times M$$

$$M=19$$

$$19= D / (1-D) \quad 19-19D=D$$

$$19=20D \quad 19/20=0.95=D$$

The discount factor D reflects both time value of money and probability of distress.

$$D=(1-d) / (1+r)$$

$$=(1-d)/1.03$$

$$0.95= (1-d) / 1.03$$

$$0.95 \times 1.03 = 1-d$$

$$.9785= 1-d$$

$$d=.0215$$

b)

I – costs of financial distress – It can be difficult and costly for distressed firms to raise capital. Issuing Principal debtholders are the customers (policyholders)

Loss reserves and unearned premium reserves are supported by funds held for payment to shareholders

The above gives debtholders more relative power in the relationship.

Insurers must maintain lower levels of risk because the relationship with debt holders is ongoing and customer relationships will suffer if their risk attitudes are ignored.

I.

Insurers are particularly vulnerable to financial distress.

Insurance payment is a promise of payment contingent upon certain future events.

A distressed firm can lose many customers which may cause it to lower prices to attract business

Customers who remain expect discounted prices and this sets into motion a vicious cycle that includes substantial loss of future earnings potential.

II.

Reinsurance is the dominant means of hedging insurance risk

Since reinsurance is recorded in financial statements the insurance industry is more transparent than others in its use of risk transfer

c)

- I. Distressed firms are able to issue only high yield bonds
Which involve substantial recurring expense in terms of making high interest payments
- II. Also, bonds often have covenants that require specific firm action in the case of financial distress which restricts a firm's flexibility in responding to the distress.